

E-tiGo Pricing Engine

How It Works & What You Can Control / September 2026

This document walks through how pricing works in the E-tiGo platform, what each setting does, and the business decisions you can make through the admin dashboard without needing any code changes.

How Fares Are Calculated

Every fare in E-tiGo is calculated using a straightforward formula. When a passenger requests a ride, the system looks up the pricing configuration for their city and chosen vehicle class, gets the distance and estimated travel time for the route, and runs this calculation:

FARE FORMULA

$$\text{Fare} = \max(\text{Minimum Fare}, \text{Base Fare} + (\text{Distance} \times \text{Per-km Rate}) + (\text{Duration} \times \text{Per-minute Rate}) + \text{Waiting Charge})$$

The **max()** part is important — it means the fare can never drop below whatever you set as the minimum fare. If a ride is very short (say, 0.5 km down the street), the calculated amount might be lower than the minimum, so the system bumps it up. This protects driver earnings on short trips.

What Each Rate Controls

RATE	WHAT IT DOES	BUSINESS IMPACT
Base Fare	Fixed amount charged on every trip, regardless of distance or time.	Sets your floor revenue per ride. Higher base = more predictable earnings per trip.
Per-km Rate	Charged per kilometre of the route.	Makes longer trips proportionally more expensive. The primary revenue lever for most rides.
Per-minute Rate	Charged per minute of estimated travel time.	Compensates drivers for time in traffic. Higher rates in congested cities make sense.

Minimum Fare	Absolute floor — fare can never go below this.	Protects driver economics on very short trips. Should cover at least fuel + opportunity cost.
Waiting Time Rate	Per-minute charge when the driver is waiting beyond the free period at pickup.	Compensates drivers for idle time. Applied after the free waiting window expires.
Free Waiting Minutes	Grace period before waiting charges begin (default: 5 minutes).	Balances passenger convenience with driver time. Industry standard is 3–5 minutes.

Example Fare Calculation

LAGOS — ECONOMY CLASS — 8.5 KM TRIP, 15 MIN ESTIMATED	
Base fare	₦500.00
Distance: 8.5 km x ₦100/km	₦850.00
Time: 15 min x ₦20/min	₦300.00
Total fare	₦1,650.00

In this example, the calculated fare (₦1,650) is above the minimum fare (say, ₦700), so the passenger pays ₦1,650. If this had been a 0.3 km trip with a 2-minute duration, the calculation would give ₦570 — below the minimum — so the fare would be ₦700 instead.

Waiting Time Charges

When a driver arrives at the pickup location and the passenger isn't ready, waiting time charges kick in — but only after a free grace period. Here's how it works:

WAITING CHARGE FORMULA

Waiting Charge = max(0, Actual Wait – Free Waiting Minutes) x Waiting Time Rate

Every pricing config includes a **free_waiting_minutes** value (defaults to 5 minutes). The driver waits without the passenger being charged anything for that period. After the free period expires, every additional minute is charged at the **waiting_time_rate**.

Example: Waiting Time on a Lagos Economy Ride

DRIVER WAITS 8 MINUTES — 5 MIN FREE, RATE: ₦15/MIN
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Total wait time	8 min
Free waiting period	–5 min
Chargeable minutes	3 min
Waiting charge: 3 min x ■15/min	■45.00

How estimates work: When passengers request a fare estimate, they see the waiting time policy (free minutes and per-minute rate) so they know the rules before booking. The actual waiting charge is only applied when the ride is completed, based on how long the driver actually waited.

Pricing by City and Vehicle Class

Pricing is configured per **city + vehicle class** combination. This means you can set completely different rates for:

- Different cities (Lagos pricing vs. Abuja pricing vs. Port Harcourt pricing)
- Different vehicle classes within the same city (Economy vs. Premium vs. SUV in Lagos)

This gives you fine-grained control. You might want Economy rides in Lagos to start at ■500, but Premium rides to start at ■1,200 — and those same classes in Abuja might have completely different numbers based on local economics.

Key point: You need to create a pricing config for every city + vehicle class combination you want to offer rides in. If a city has 4 active vehicle classes, that's 4 pricing configs to set up. Without a pricing config, that vehicle class won't appear in fare estimates for that city.

Versioning — How Pricing Changes Work

Pricing configs are **versioned and append-only**. When you create a new pricing config for the same city + vehicle class, the system doesn't overwrite the old one — it creates a new version alongside it. This is deliberate, and it gives you two important capabilities:

Price History

You can always look back and see what your rates were at any point in time. This is useful for auditing, resolving disputes, and understanding revenue trends. The admin dashboard shows you all versions with their effective dates.

Scheduled Price Changes

Every pricing config has an **effective_from** date. You can create a new pricing config today that won't kick in until next Monday. The system will keep using the current rates until that date, then automatically switch to the new ones. No manual intervention needed on the day.

In-progress ride protection: When a passenger books a ride, the system takes a snapshot of the pricing at that moment and locks it to that ride. If you change pricing while someone is mid-trip, their fare stays on the rate they were quoted. This prevents disputes and builds passenger trust.

Business Decisions You Can Make

Here are the main levers you have through the admin dashboard, and when you might want to pull each one:

Adjust Rates by City

If a particular city has higher fuel costs, more traffic, or a different cost of living, you can set rates accordingly. You might charge more per km in a city with expensive fuel and more per minute in a city with heavy congestion.

Differentiate Vehicle Classes

Premium and luxury classes should carry a meaningful price premium. A rule of thumb: Premium is typically 1.5–2x Economy, SUV is 2–2.5x. EV classes could be priced at a slight discount to incentivise green rides, or at a premium if positioned as a luxury offering.

Schedule Rate Changes

Planning a rate increase? Create the new pricing config with a future effective date. Want to run a promotional week with lower fares? Set lower rates effective Monday and higher rates effective the following Monday. Both will apply automatically.

Set Minimum Fares Strategically

The minimum fare is your most important tool for driver retention. If it's too low, drivers lose money on short trips and stop accepting them. If it's too high, passengers walk instead of booking. Check your average trip distance — minimum fare should be attractive enough that drivers don't skip short rides.

Configure Waiting Time Policy

You control both the free waiting period and the per-minute rate. The default is 5 minutes free (Bolt uses 4). Setting this too low frustrates passengers; too high and drivers lose money waiting around. You can adjust per city and vehicle class.

Launch a New City

When expanding to a new city: create the city in the admin panel, enable the vehicle classes you want to offer there, then create pricing configs for each city + vehicle class combination. The city goes live for passengers as soon as pricing is in place.

A/B Test Pricing (Manual)

While we don't have automated A/B testing built in yet, you can compare performance across cities with different rate structures. Set different rates in two similar cities for the same vehicle class, then compare ride volumes and completion rates in the admin reports to see which pricing performs better.

What's Coming Next

The pricing engine is built to be extended. Here's what we're planning for future phases:

FEATURE	WHAT IT WILL DO	TIMELINE
Surge Pricing	Dynamic multiplier when demand exceeds supply in an area. You'll control the max multiplier cap and the thresholds.	Phase 2
Promo Integration	Promotional discounts applied on top of the calculated fare. The promo engine is a separate module that hooks into fare calculation.	Phase 1 (next sprint)
Tier Discounts	Automatic booking fee discounts for passengers in higher loyalty tiers. Configured through the gamification settings.	Phase 1
Real Maps Integration	Currently using straight-line distance estimates with a road-distance adjustment. Once we confirm the maps provider, fare estimates will use actual road distances and real-time traffic data.	Pending provider selection

Quick Reference

Admin Actions

ACTION	WHERE IN DASHBOARD
Create new pricing	Admin → Pricing → New Config
View all pricing history	Admin → Pricing (filter by city / vehicle class)
Check current active rate	Admin → Pricing → Current (select city + class)
Schedule a future rate	Create new pricing with a future "Effective From" date

E-tiGo Backend — Pricing Engine Documentation

Questions? Reach out and I'll walk you through any of this in more detail.